TP03

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Link: https://github.com/Mookneii/Data_Structure_And_Algorithms.git

1.

```cpp
#include <iostream>
using namespace std;
int main(){
    int n1, n2, n3;
    int *p1 = &n1;
    int *p2 = &n2;
    int *p3 = &n3;
    cout<<"n1 address at "<<p1<<", value "<<*p1<<endl;
    cout<<"n2 address at "<<p2<<", value "<<*p2<<endl;
    cout<<"n3 address at "<<p3<<", value "<<*p3<<endl;
       *p3 = *p1 + *p2;
    cout<<"After update: \n";
    cout<<"n3 address at "<<p3<<", value "<<*p3<<endl;
    return 0;
}
```

```
n1 address at 0x685fdffcd4, value 32767
n2 address at 0x685fdffcd0, value -702968288
n3 address at 0x685fdffccc, value 0
After update:
n3 address at 0x685fdffccc, value -702935521
```

2.

```cpp
#include <iostream>
using namespace std;
int main(){
    int num;
    cout<<"Enter n :";cin>>num;
    int *n = &num;
    cout<<"Modify by n + 7: "<<*n+7<<endl;
    return 0;
}
```

```
Enter n :6
Modify by n + 7: 13
```

3.

```cpp
#include <iostream>
using namespace std;
void exchange(float *a, float *b){
    float temp = *a;
    *a = *b;
    *b = temp;
}
int main(){
    float a=5;
    float b=6;

    cout<<"Before modifier a = "<<a<<", b = "<<b<<endl;
    exchange(&a, &b);
    cout<<"Before modifier a = "<<a<<", b = "<<b<<endl;
    return  0;
}
```

```
Before modifier a = 5, b = 6
Before modifier a = 6, b = 5
```

4.

```cpp
#include <iostream>
#include<cmath>
using namespace std;
void solveEquation(int a, int b, int c, float *x1, float *x2, float *delta){
    *delta = b*b -4*a*c;
    if( *delta > 0){
        *x1 = (-b -sqrt(*delta))/2*a;
        *x1 = (-b +sqrt(*delta))/2*a;
    }
    else if(*delta ==0){
        *x1 = *x2 = -b / 2*a;
    }
    else{
        cout<<"No root";
    }
}
int main(){
    int a= 1;
    int b = 8;
    int c = 16;
    float x1, x2, delta;
    solveEquation(a,b,c,&x1,&x2,&delta);
    cout <<"delta : "<<delta<<endl;
    if(delta >0){
        cout<<"x1 = "<<x1<<",x2 = "<<x2<<endl;
    }else if(delta == 0){
        cout<<"x1 =x2= "<<x1<<endl;
    }
    else{
        cout<<"No root"<<endl;
    }
}
```

```
delta : 0
x1 =x2= -4
```

5.

```cpp
#include <iostream>
using namespace std;

void findMaxMin(int* arr, int* max, int* min) {
    *max = *arr;
    *min = *arr;

    for (int i = 1; i < 7; i++) {
        if (*(arr + i) > *max) {
            *max = *(arr + i);
        }
        if (*(arr + i) < *min) {
            *min = *(arr + i);
        }
    }
}
int main() {
    int arr[7];

    cout << "Enter 7 integer numbers:\n";


    for (int i = 0; i < 7; i++) {
        cin >> *(arr + i);
    }

    int max, min;


    findMaxMin(arr, &max, &min);

    cout << "Maximum value: " << max << endl;
    cout << "Minimum value: " << min << endl;

    return 0;
}
```

```
Enter 7 integer numbers:
21
2
3
6
43
7
6
Maximum value: 43
Minimum value: 2
PS D:\Data_Structure_and_Programming\Lab3>
```

6.

```cpp
#include <iostream>
using namespace std;

int main() {
    float arr[7];
    float* ptr = arr;
    cout << "Enter 7 float numbers:\n";
    for (int i = 0; i < 7; i++) {
        cin >> *(ptr + i);
    }
    float* displayPtr = arr;

    cout << "\nNumbers in array:\n";
    for (int i = 0; i < 7; i++) {
        cout << *(displayPtr + i) << " ";
    }


    float sum = 0;
    float product = 1;

    float* calcPtr = arr;

    for (int i = 0; i < 7; i++) {
        sum += *(calcPtr + i);
        product *= *(calcPtr + i);
    }

    cout << "\n\nSum = " << sum << endl;
    cout << "Product = " << product << endl;

    return 0;
}
```

```
Enter 7 float numbers:
2.12
4.3212
54.32223
1.23244
5.3223
5.322323
6.32323

Numbers in array:
2.12 4.3212 54.3222 1.23244 5.3223 5.32232 6.32323

Sum = 78.9637
Product = 109856
```

7.

```cpp
#include <iostream>
using namespace std;
void sum1(double *sum, int n) {
    *sum = 0;

    for (int i = 1; i <= n; i++) {
        *sum += 1.0 / (i * i);
    }
}
int main() {
    double result;
    int n;

    cout << "Enter a positive integer n: ";
    cin >> n;


    sum1(&result, n);

    cout << "Sum = " << result << endl;

    return 0;
}
```

```
Enter a positive integer n: 23
Sum = 1.60239
```